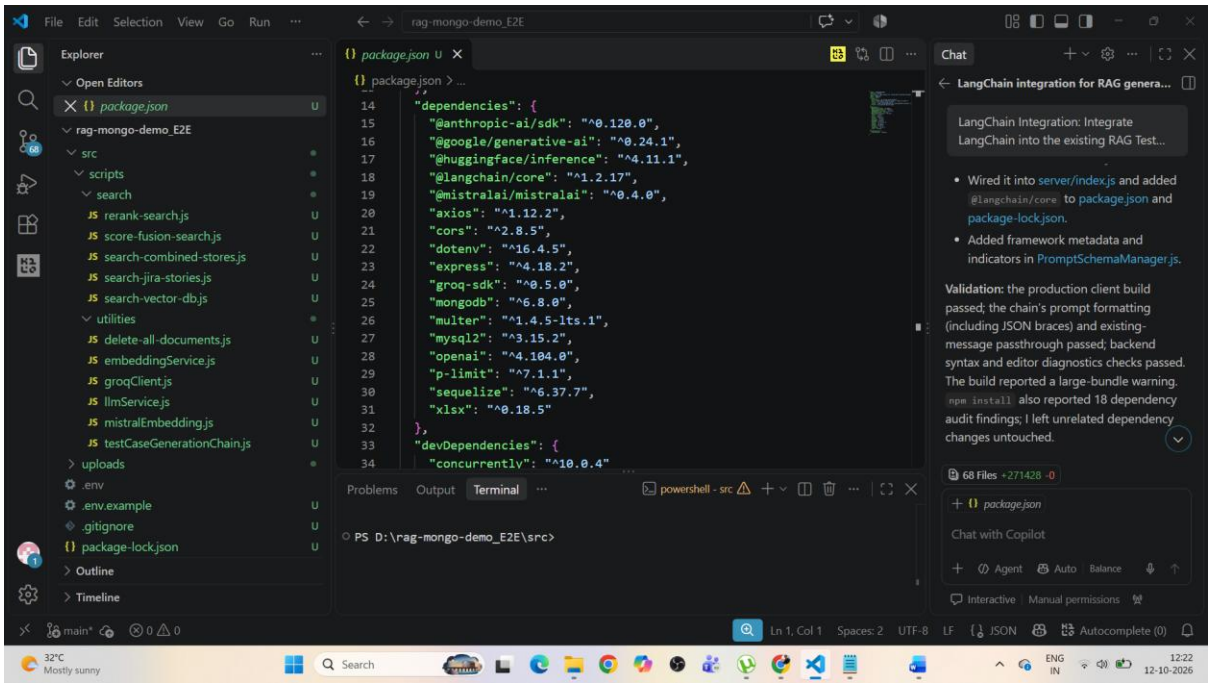
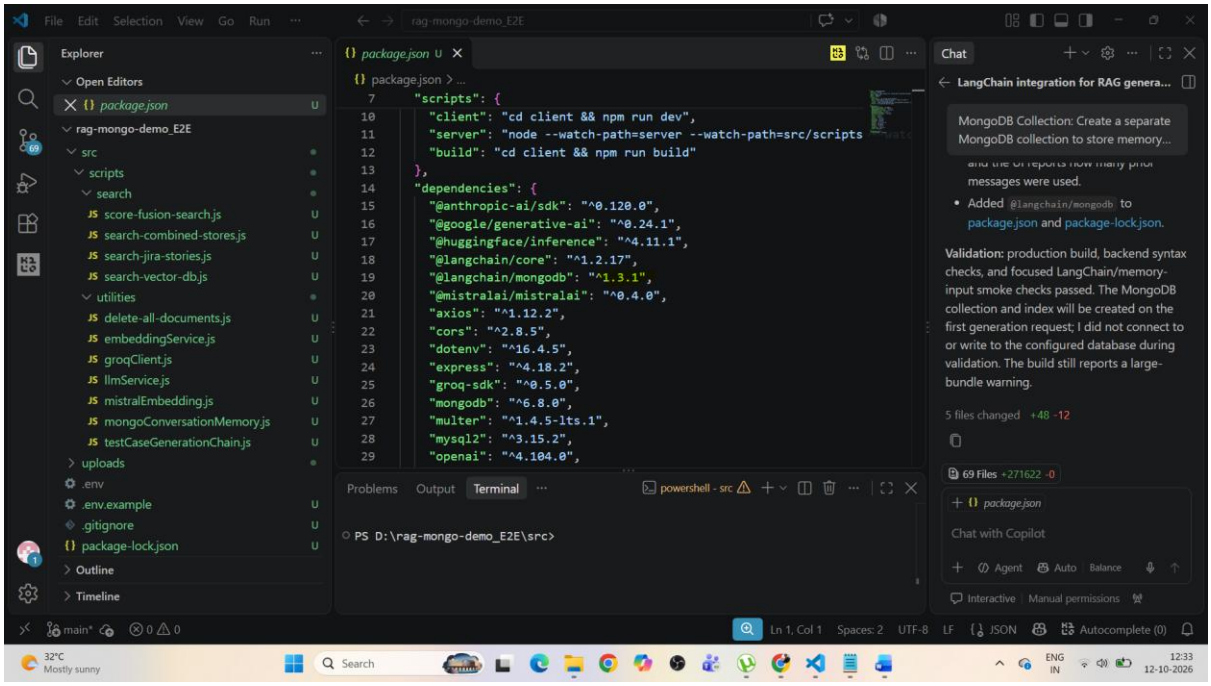
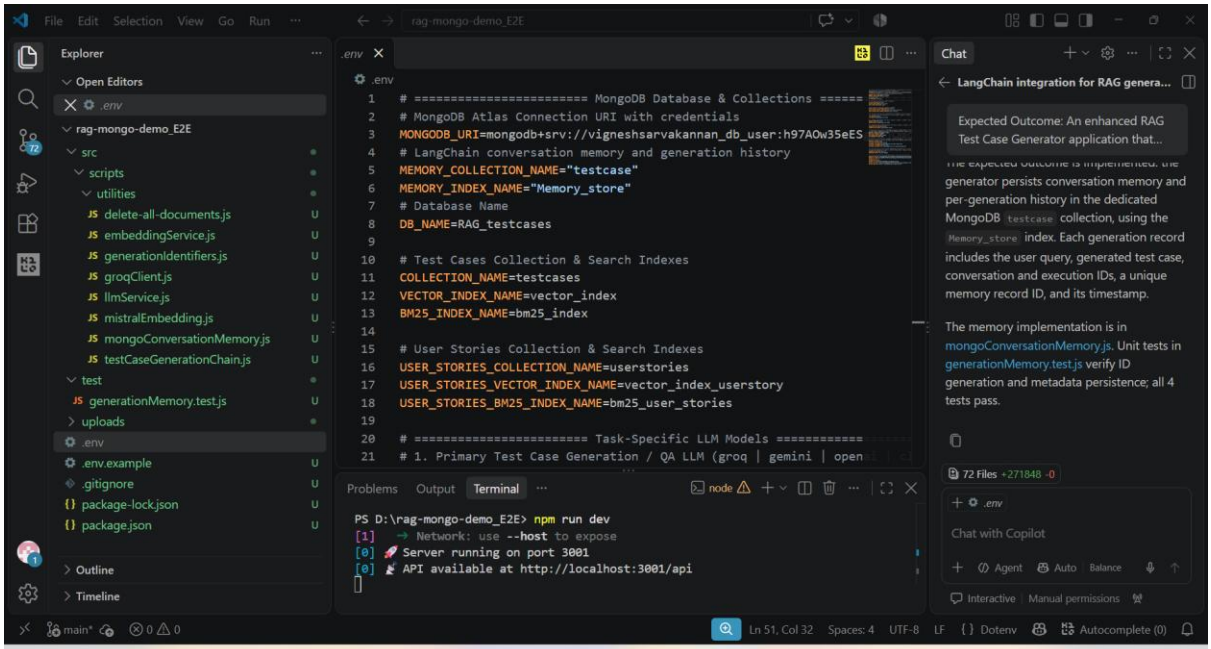
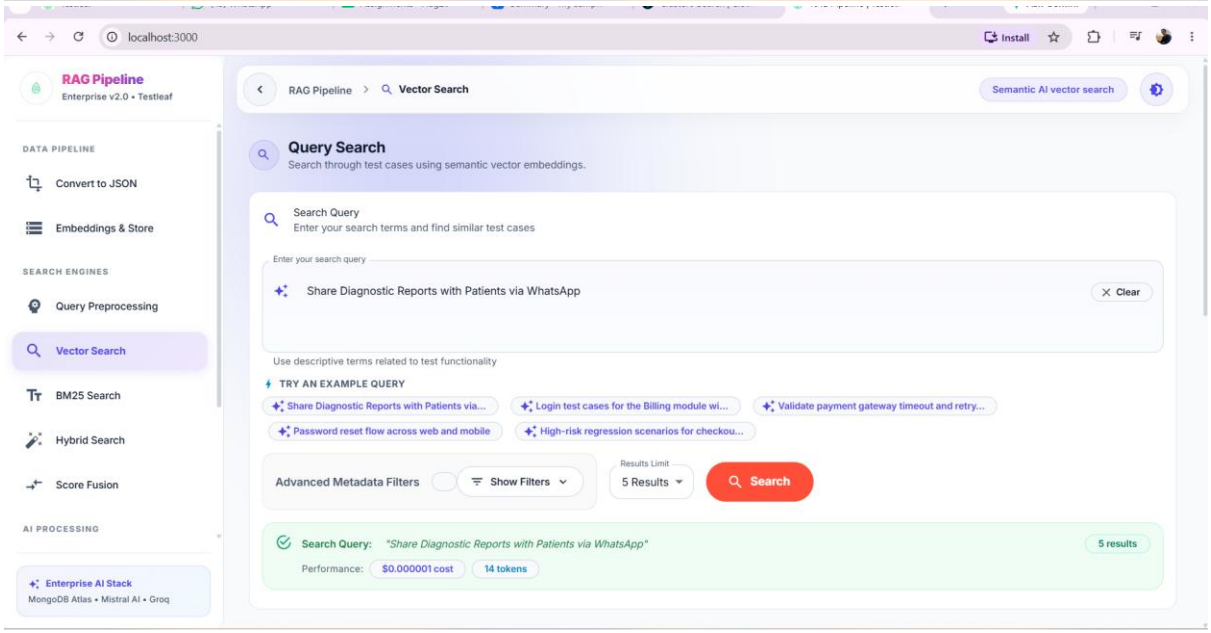
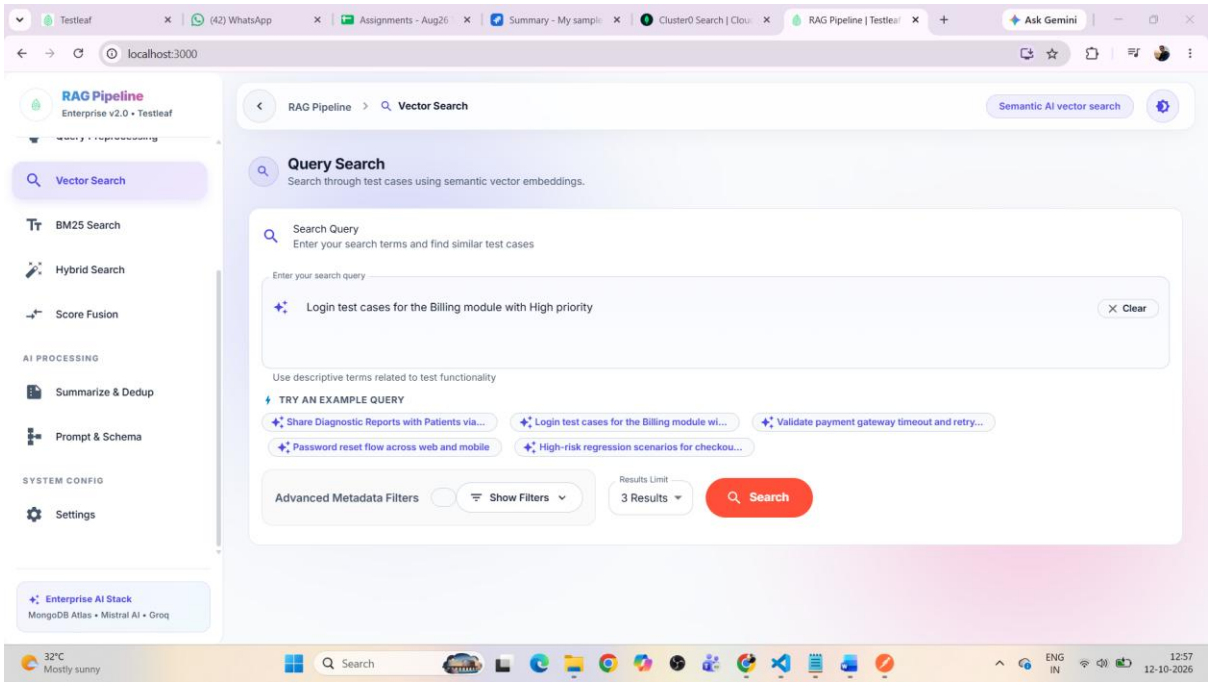
Adding the memory to RAG enables test cases

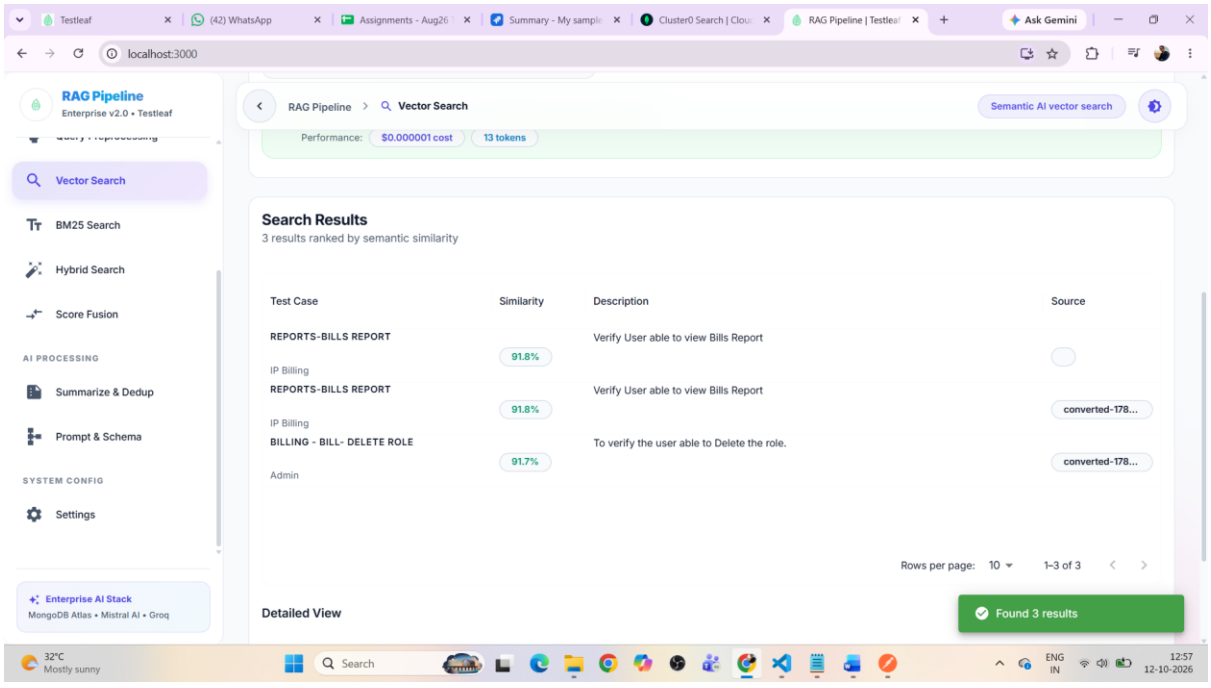


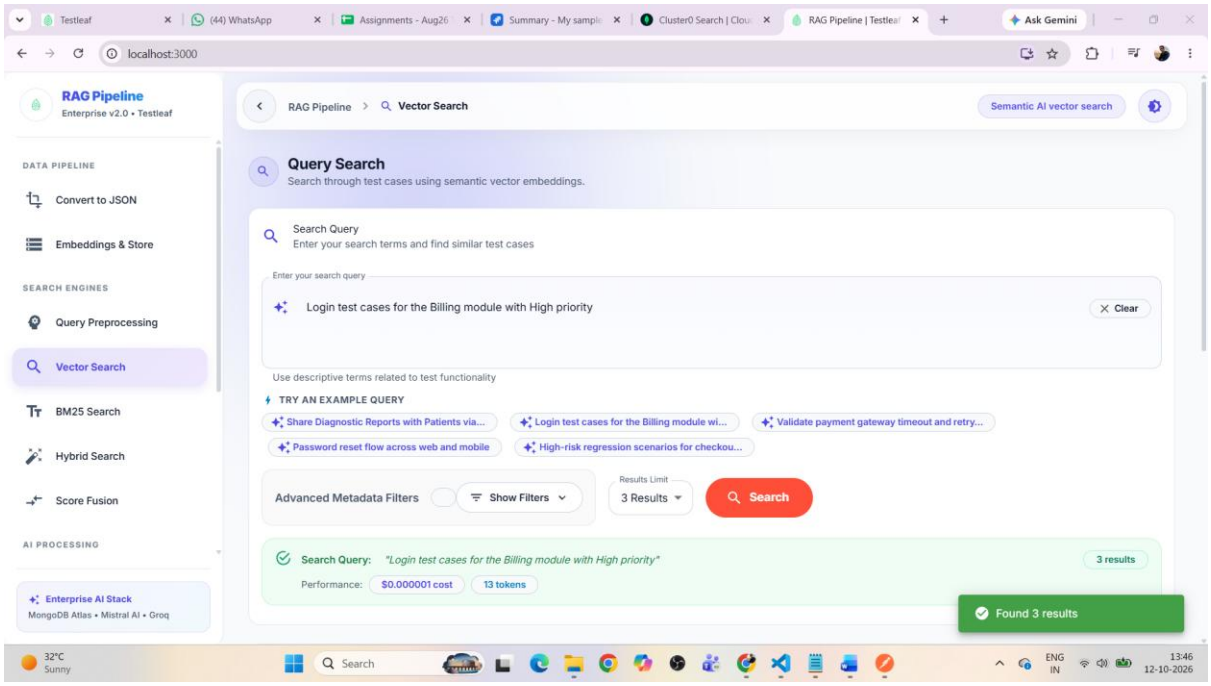


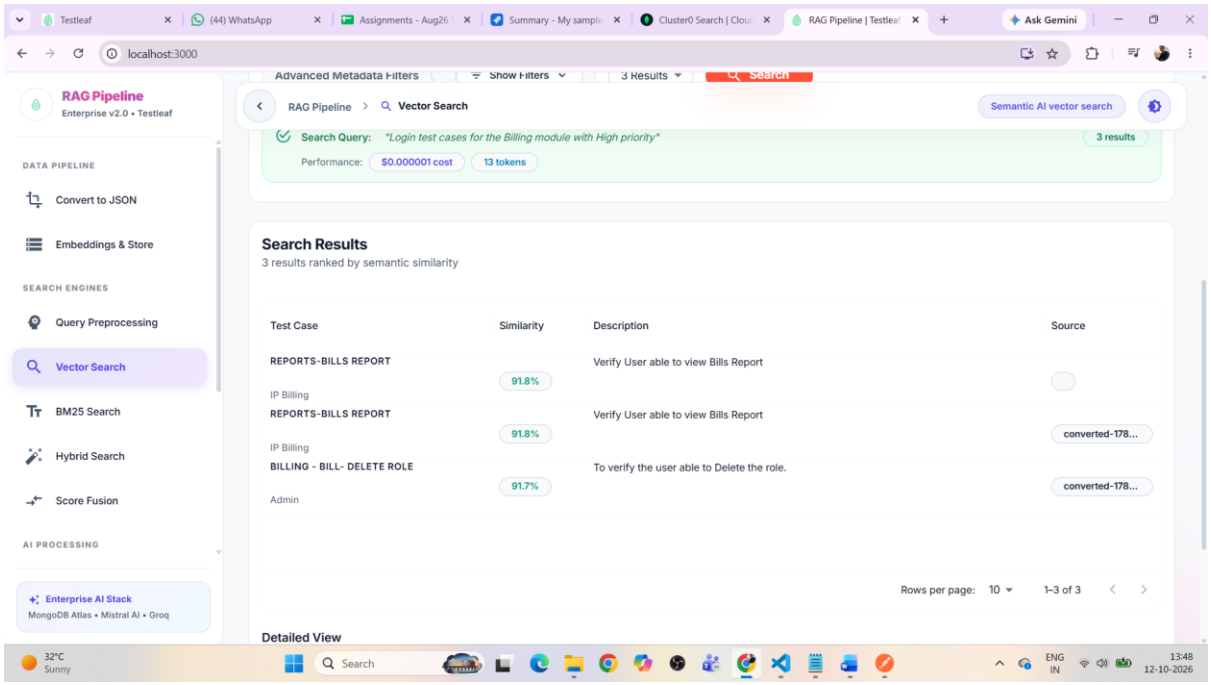

File Edit Selection View Go Run
rag-mongo-demo_E2E
Explorer
Open Editors
.env
rag-mongo-demo_E2E
src
scripts
utilities
delete-all-documents.js
embeddingService.js
generationIdentifiers.js
groqClient.js
llmService.js
mistralEmbedding.js
mongoConversationMemory.js
testCaseGenerationChain.js
test
generationMemory.test.js
uploads
.env
.env.example
.gitignore
package-lock.json
package.json
Outline
Timeline
.env
1 # ======================== MongoDB Database & Collections ======
2 # MongoDB Atlas Connection URI with credentials
3 MONGODB_URI=mongodb+srv://vigneshsarvakannan_db_user:h97AOw35eES
4 # LangChain conversation memory and generation history
5 MEMORY_COLLECTION_NAME="testcase"
6 MEMORY_INDEX_NAME="Memory_store"
7 # Database Name
8 DB_NAME=RAG_testcases
9
10 # Test Cases Collection & Search Indexes
11 COLLECTION_NAME=testcases
12 VECTOR_INDEX_NAME=vector_index
13 BM25_INDEX_NAME=bm25_index
14
15 # User Stories Collection & Search Indexes
16 USER_STORIES_COLLECTION_NAME=userstories
17 USER_STORIES_VECTOR_INDEX_NAME=vector_index_userstory
18 USER_STORIES_BM25_INDEX_NAME=bm25_user_stories
19
20 # ======================== Task-Specific LLM Models ============
21 # 1. Primary Test Case Generation / QA LLM (groq | gemini | open
Problems Output Terminal
node
PS D:\rag-mongo-demo_E2E> npm run dev
[1] → Network: use --host to expose
[0] Server running on port 3001
[0] API available at http://localhost:3001/api
Chat
LangChain integration for RAG genera...
Expected Outcome: An enhanced RAG Test Case Generator application that...
generator persists conversation memory and per-generation history in the dedicated MongoDB testcase collection, using the Memory_store index. Each generation record includes the user query, generated test case, conversation and execution IDs, a unique memory record ID, and its timestamp.
The memory implementation is in mongoConversationMemory.js. Unit tests in generationMemory.test.js verify ID generation and metadata persistence; all 4 tests pass.
72 Files +271848 -0
.env
Chat with Copilot
Agent Auto Balance
Interactive Manual permissions
main* 0 0
Ln 51, Col 32 Spaces: 4 UTF-8 LF Dotenv Autocomplete (0)
32°C Mostly sunny
Search
ENG IN
12:49 12-10-2026

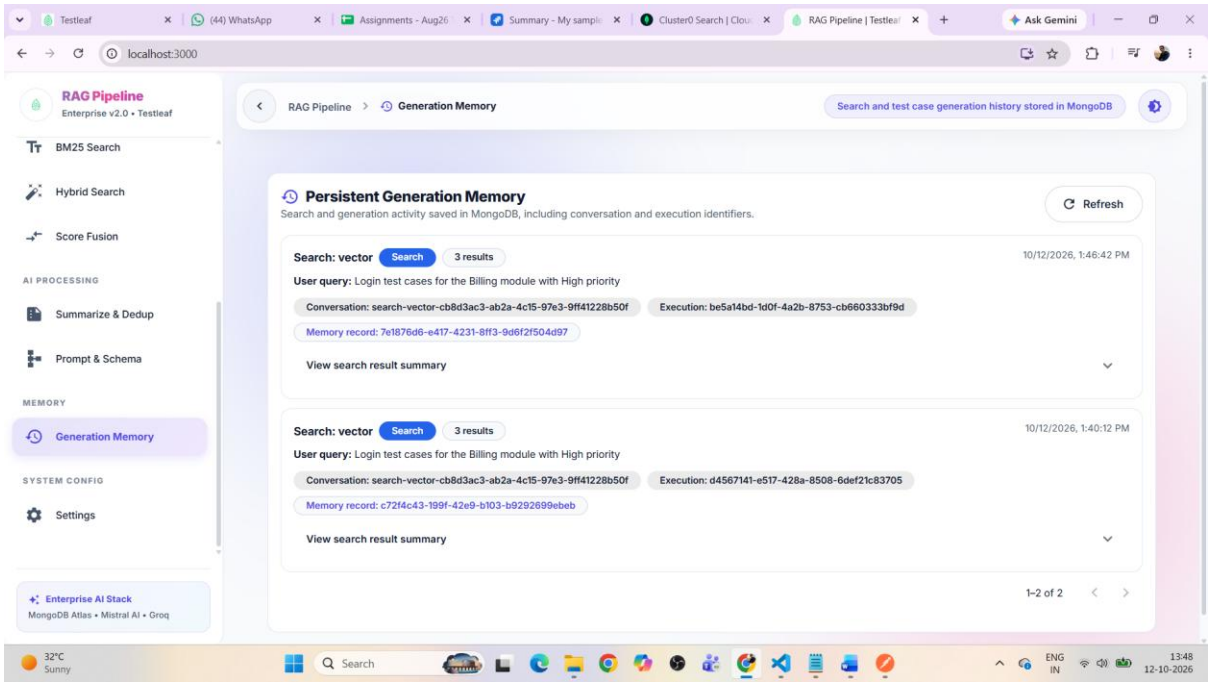

localhost:3000
Install
RAG Pipeline
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SEARCH ENGINES
Query Preprocessing
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BM25 Search
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MongoDB Atlas • Mistral AI • Groq
RAG Pipeline > Vector Search
Semantic AI vector search
Query Search
Search through test cases using semantic vector embeddings.
Search Query
Enter your search terms and find similar test cases
Enter your search query
Share Diagnostic Reports with Patients via WhatsApp
Clear
Use descriptive terms related to test functionality
TRY AN EXAMPLE QUERY
Share Diagnostic Reports with Patients via...
Login test cases for the Billing module wi...
Validate payment gateway timeout and retry...
Password reset flow across web and mobile
High-risk regression scenarios for checkou...
Advanced Metadata Filters
Show Filters
Results Limit
5 Results
Search
Search Query: "Share Diagnostic Reports with Patients via WhatsApp"
5 results
Performance: $0.000001 cost 14 tokens

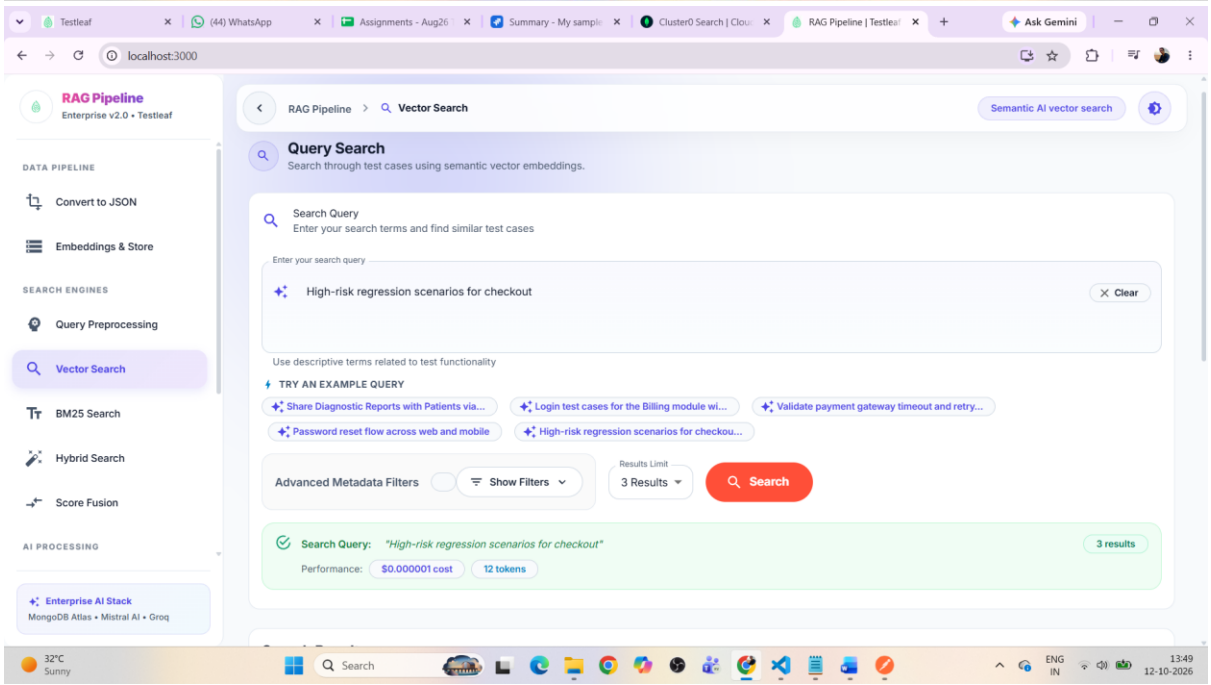

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RAG Pipeline
Vector Search
Semantic AI vector search
Query Search
Search through test cases using semantic vector embeddings.
Search Query
Enter your search terms and find similar test cases
Enter your search query
Login test cases for the Billing module with High priority
Clear
Use descriptive terms related to test functionality
TRY AN EXAMPLE QUERY
Share Diagnostic Reports with Patients via...
Login test cases for the Billing module wi...
Validate payment gateway timeout and retry...
Password reset flow across web and mobile
High-risk regression scenarios for checkou...
Advanced Metadata Filters
Show Filters
Results Limit
3 Results
Search

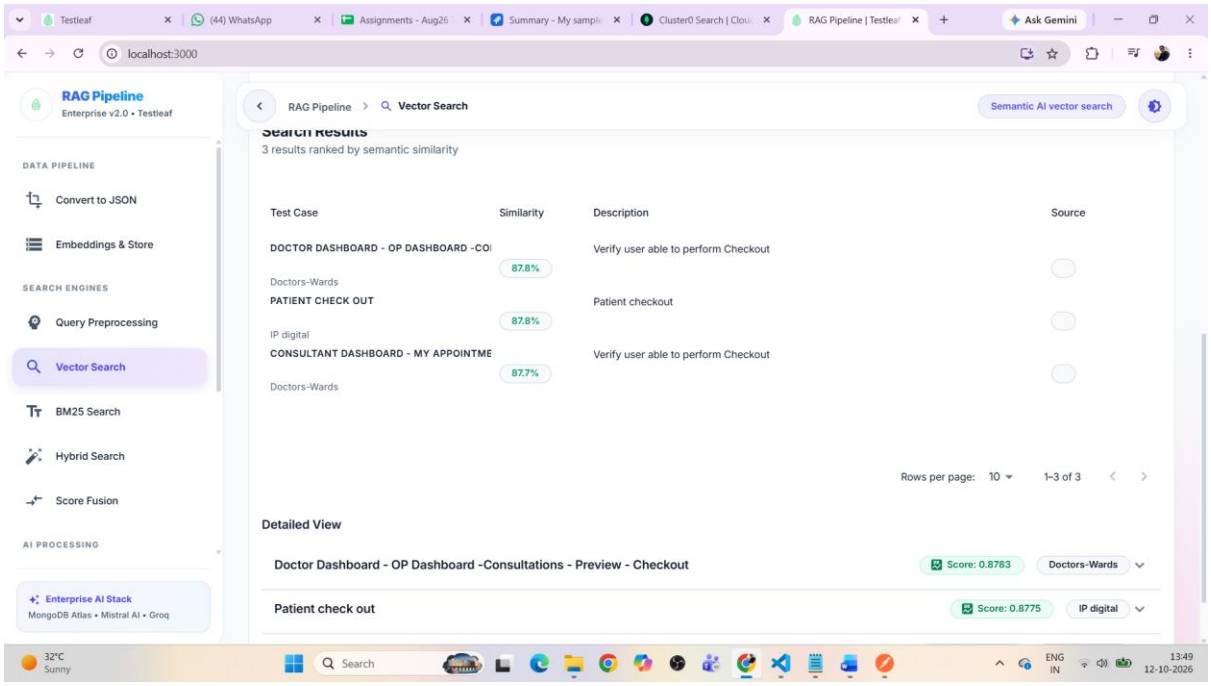

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RAG Pipeline
Vector Search
Semantic AI vector search
Performance:
$0.000001 cost
13 tokens
Search Results
3 results ranked by semantic similarity
Test Case
Similarity
Description
Source
REPORTS-BILLS REPORT
IP Billing
91.8%
Verify User able to view Bills Report
REPORTS-BILLS REPORT
IP Billing
91.8%
Verify User able to view Bills Report
converted-178...
BILLING - BILL- DELETE ROLE
Admin
91.7%
To verify the user able to Delete the role.
converted-178...
Rows per page: 10
1–3 of 3
Detailed View
Found 3 results

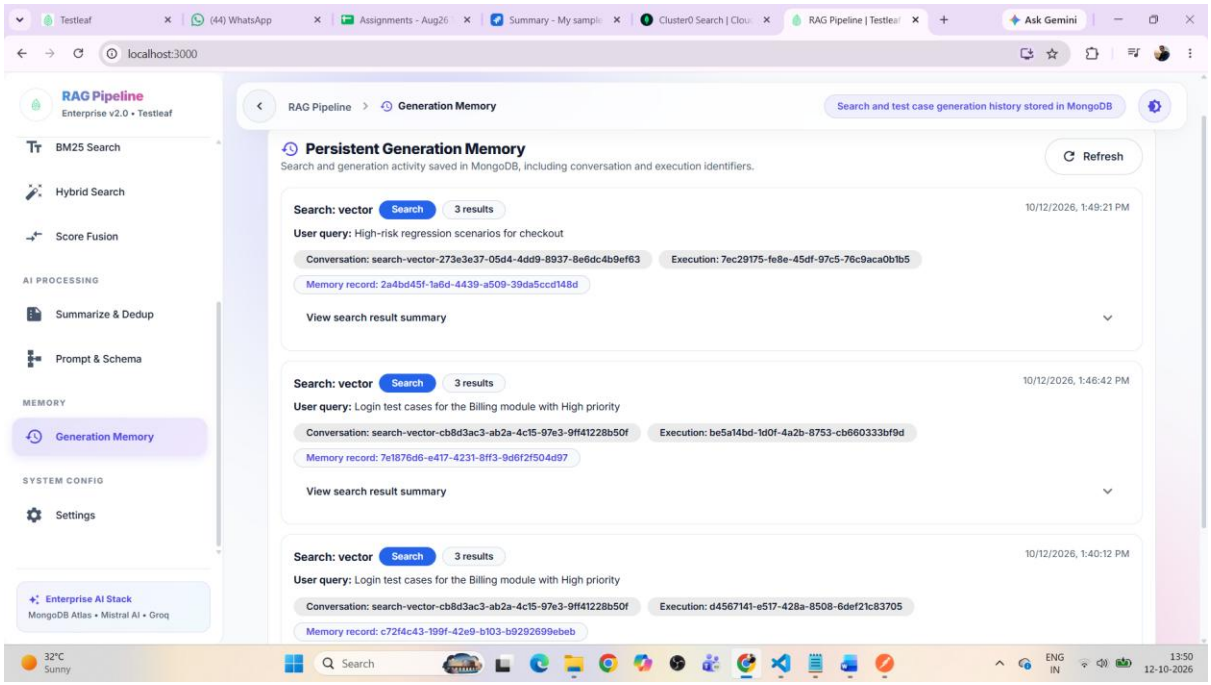

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Show Filters
Results Limit
3 Results
Search
Search Query: "Login test cases for the Billing module with High priority"
3 results
Performance: $0.000001 cost
13 tokens
Found 3 results

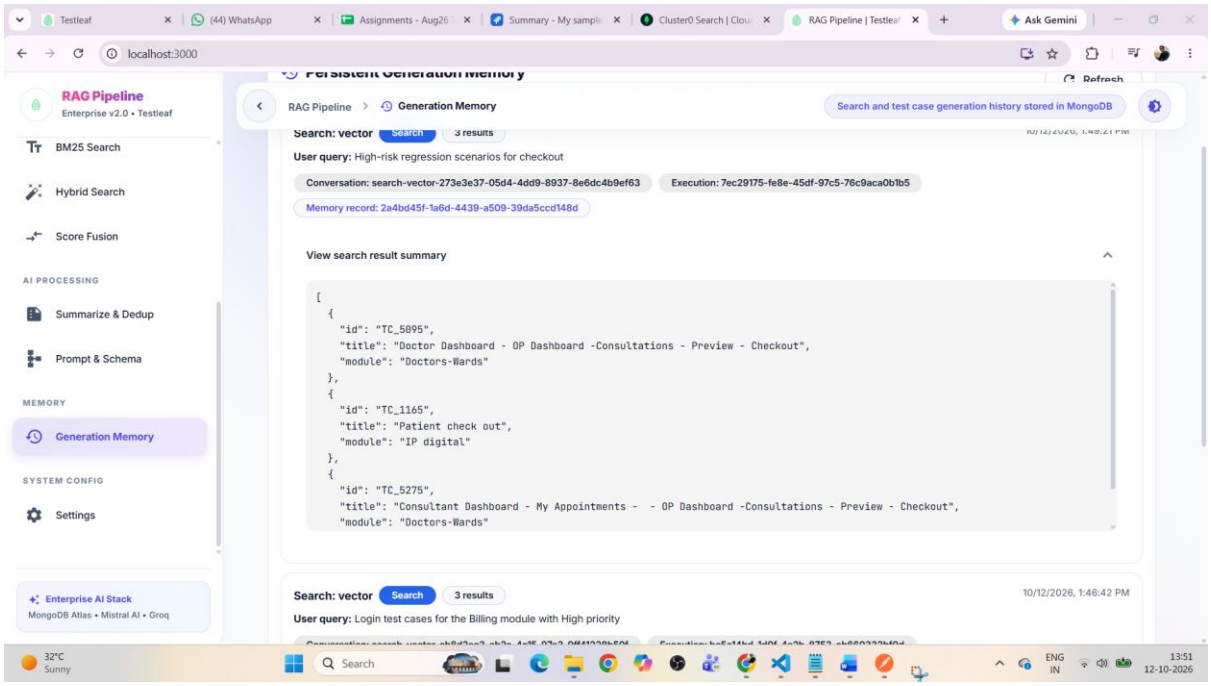

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Search Query: "Login test cases for the Billing module with High priority"
3 results
Performance: $0.000001 cost
13 tokens
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3 results ranked by semantic similarity
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Similarity
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91.8%
Verify User able to view Bills Report
REPORTS-BILLS REPORT
IP Billing
91.8%
Verify User able to view Bills Report
converted-178...
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91.7%
To verify the user able to Delete the role.
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Rows per page: 10
1–3 of 3
Detailed View

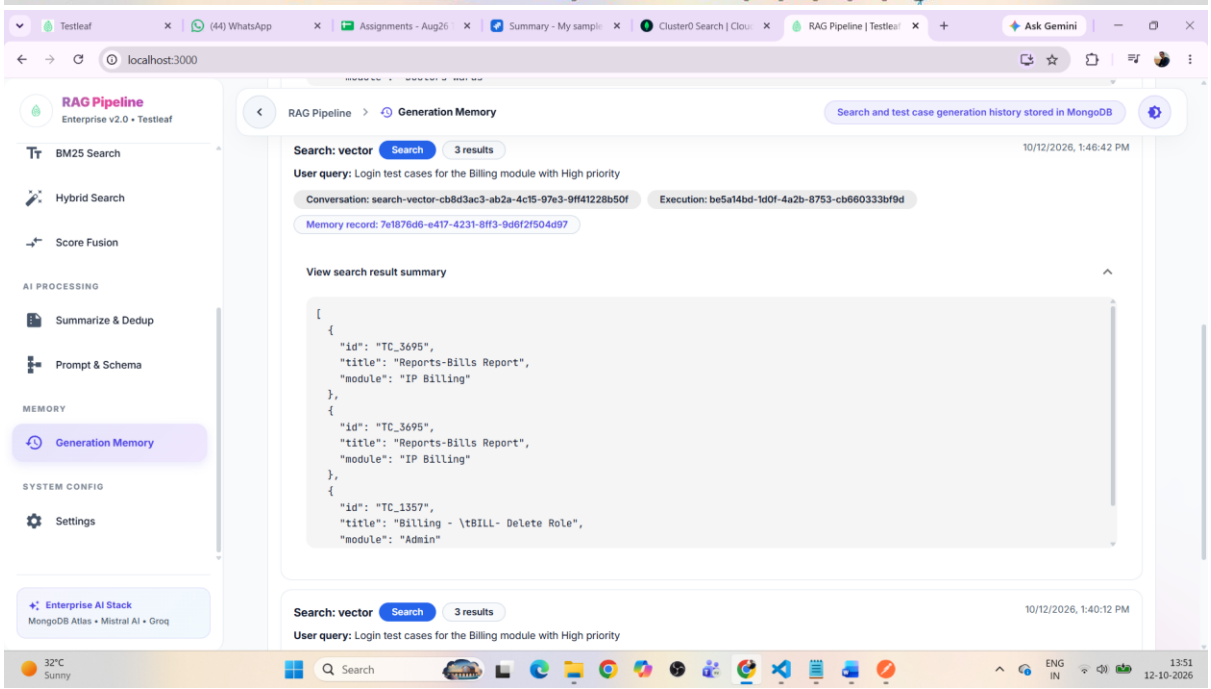

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MEMORY
Generation Memory
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RAG Pipeline
Generation Memory
Search and test case generation history stored in MongoDB
Persistent Generation Memory
Search and generation activity saved in MongoDB, including conversation and execution identifiers.
Refresh
Search: vector
Search
3 results
10/12/2026, 1:46:42 PM
User query: Login test cases for the Billing module with High priority
Conversation: search-vector-cb8d3ac3-ab2a-4c15-97e3-9ff41228b50f
Execution: be5a14bd-1d0f-4a2b-8753-cb660333bf9d
Memory record: 7e1876d6-e417-4231-8ff3-9d6f2f504d97
View search result summary
Search: vector
Search
3 results
10/12/2026, 1:40:12 PM
User query: Login test cases for the Billing module with High priority
Conversation: search-vector-cb8d3ac3-ab2a-4c15-97e3-9ff41228b50f
Execution: d4567141-e517-428a-8508-6def21c83705
Memory record: c72f4c43-199f-42e9-b103-b9292699ebeb
View search result summary
1–2 of 2
32°C
Sunny
Search
ENG
IN
13:48
12-10-2026


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High-risk regression scenarios for checkou...
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Show Filters
Results Limit
3 Results
Search
Search Query: "High-risk regression scenarios for checkout"
3 results
Performance: $0.000001 cost
12 tokens
32°C
Sunny
Search
ENG
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Search Results
3 results ranked by semantic similarity
Test Case
Similarity
Description
Source
DOCTOR DASHBOARD - OP DASHBOARD -CO
Verify user able to perform Checkout
87.8%
Doctors-Wards
PATIENT CHECK OUT
Patient checkout
87.8%
IP digital
CONSULTANT DASHBOARD - MY APPOINTME
Verify user able to perform Checkout
87.7%
Doctors-Wards
Rows per page: 10
1–3 of 3
Detailed View
Doctor Dashboard - OP Dashboard -Consultations - Preview - Checkout
Score: 0.8783
Doctors-Wards
Patient check out
Score: 0.8775
IP digital


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Persistent Generation Memory
Search and generation activity saved in MongoDB, including conversation and execution identifiers.
Refresh
Search: vector
Search
3 results
10/12/2026, 1:49:21 PM
User query: High-risk regression scenarios for checkout
Conversation: search-vector-273e3e37-05d4-4dd9-8937-8e6dc4b9ef63
Execution: 7ec29175-fe8e-45df-97c5-76c9aca0b1b5
Memory record: 2a4bd45f-1a6d-4439-a509-39da5ccd148d
View search result summary
Search: vector
Search
3 results
10/12/2026, 1:46:42 PM
User query: Login test cases for the Billing module with High priority
Conversation: search-vector-cb8d3ac3-ab2a-4c15-97e3-9ff41228b50f
Execution: be5a14bd-1d0f-4a2b-8753-cb660333bf9d
Memory record: 7e1876d6-e417-4231-8ff3-9d6f2f504d97
View search result summary
Search: vector
Search
3 results
10/12/2026, 1:40:12 PM
User query: Login test cases for the Billing module with High priority
Conversation: search-vector-cb8d3ac3-ab2a-4c15-97e3-9ff41228b50f
Execution: d4567141-e517-428a-8508-6def21c83705
Memory record: c72f4c43-199f-42e9-b103-b9292699ebeb


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RAG Pipeline > Generation Memory
Search and test case generation history stored in MongoDB
Search: vector Search 3 results
User query: High-risk regression scenarios for checkout
Conversation: search-vector-273e3e37-05d4-4dd9-8937-8e6dc4b9ef63
Execution: 7ec29175-fe8e-45df-97c5-76c9aca0b1b5
Memory record: 2a4bd45f-1a6d-4439-a509-39da5ccd148d
View search result summary
[
{
"id": "TC_5095",
"title": "Doctor Dashboard - OP Dashboard -Consultations - Preview - Checkout",
"module": "Doctors-Wards"
},
{
"id": "TC_1165",
"title": "Patient check out",
"module": "IP digital"
},
{
"id": "TC_5275",
"title": "Consultant Dashboard - My Appointments - - OP Dashboard -Consultations - Preview - Checkout",
"module": "Doctors-Wards"
Search: vector Search 3 results
10/12/2026, 1:46:42 PM
User query: Login test cases for the Billing module with High priority


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RAG Pipeline > Generation Memory
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Search: vector Search 3 results
10/12/2026, 1:46:42 PM
User query: Login test cases for the Billing module with High priority
Conversation: search-vector-cb8d3ac3-ab2a-4c15-97e3-9ff41228b50f
Execution: be5a14bd-1d0f-4a2b-8753-cb660333bf9d
Memory record: 7e1876d6-e417-4231-8ff3-9d6f2f504d97
View search result summary
[
{
"id": "TC_3695",
"title": "Reports-Bills Report",
"module": "IP Billing"
},
{
"id": "TC_3695",
"title": "Reports-Bills Report",
"module": "IP Billing"
},
{
"id": "TC_1357",
"title": "Billing - \tBILL- Delete Role",
"module": "Admin"
Search: vector Search 3 results
10/12/2026, 1:40:12 PM
User query: Login test cases for the Billing module with High priority